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$CaO + SiO_2 \rightarrow CaSiO_3$

$6CaO + P_4O_{10} \rightarrow 2Ca_3(PO_4)_2$

Uses

Calcium oxide is used

(i) to manufacture cement, mortar and glass.

(ii) in the manufacture of sodium carbonate and slaked lime.

(iii) in the purification of sugar.

(iv) as a drying agent.

5.6.2 Calcium hydroxide

Preparation

Calcium hydroxide is prepared by adding water to quick lime, CaO.

Properties

It is a white powder. It is sparingly soluble in water. The aqueous solution is known as lime water and a suspension of slaked lime in water is known as milk of lime.

When carbon dioxide is passed through lime water, it turns milky due to the formation of calcium carbonate.

$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$

On passing excess of carbon dioxide, the precipitate dissolves to form calcium hydrogen carbonate.

$CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$

Milk of lime reacts with chlorine to form hypochlorite, a constituent of bleaching powder.

$2Ca(OH)_2 + 2Cl_2 \rightarrow$

$CaCl_2 + Ca(OCl)_2 + 2H_2O$

Uses:

Calcium hydroxide is used

(i) in the preparation of mortar, a building material.

ii) in white wash due to its disinfectant nature.

(iii) in glass making, in tanning industry, in the preparation of bleaching powder and for the purification of sugar.

5.6.3 Gypsum ($CaSO_4.2H_2O$)

Gypsum beds were formed due to the evaporation of water from the massive prehistoric sea basins. When water evaporates, the minerals present in it become concentrated, and crystallise.

Figure 5.11 A Gypsum Quarry

Properties of Gypsum

- Gypsum is a soft mineral, which is moderately soluble in water. The solubility of this mineral in water is